

Practice Quiz 03A — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 03

Nested quantifiers and negating quantified statements.

1. Consider

$$(i) \forall x \in \mathbf{Z} \exists y \in \mathbf{Z} (y > x) \quad (ii) \exists y \in \mathbf{Z} \forall x \in \mathbf{Z} (y > x).$$

(a) State each in plain English and say whether it is true or false.

Solution Statement (i) says that every integer has a larger integer. It is true: for a given x , take $y = x + 1$. Statement (ii) says that there is one integer larger than every integer. It is false. ■

(b) Negate (ii), with no negation symbol to the left of a quantifier.

Solution Its negation is

$$\forall y \in \mathbf{Z} \exists x \in \mathbf{Z} (y \leq x).$$

Indeed, $\neg(y > x)$ is $y \leq x$. ■